YAKEEN

NEET 2.0 2027

Physics | Lecture - 13

Basic Maths and Calculus

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Todays Goal

Avg value & log

Sanghas Assignment - 384

Class PPT attached

Solution upkar

Copy

$\int_{\alpha}^{\gamma} \frac{Gm_1m_2}{x^2}\,dx = Gm_1m_2 \int_{\alpha}^{\gamma} \frac{1}{x^2}\,dx = Gm_1m_2 \left[-\frac{1}{x}\right]_{\alpha}^{\gamma}$

Question gy

$= -Gm_1m_2 \left[\frac{1}{\gamma} - \frac{1}{\alpha}\right]$

$= -\frac{Gm_1m_2}{\gamma}$

$\int \frac{1}{x^2}\,dx = \int x^{-2}\,dx$

$= \frac{x^{-2+1}}{-2+1}$

$= \frac{x^{-1}}{-1}$

$= -\frac{1}{x}$ ✓

Question copy

$$\int_0^2 (4-3x)^2 dx = \text{??}$$

$$= \frac{(4-3x)^{2+1}}{(2+1)\times(-3)}$$

$$= \frac{(4-3x)^3}{-9} = -\frac{1}{9}\left[(4-3x)^3\right]_0^2$$

$$= -\frac{1}{9}\left[(-2)^3 - 4^3\right]$$

$$= -\frac{1}{9}\left[-8-64\right]$$

$$= -\frac{1}{9}\left[-72\right]$$

$$= +8$$

Question

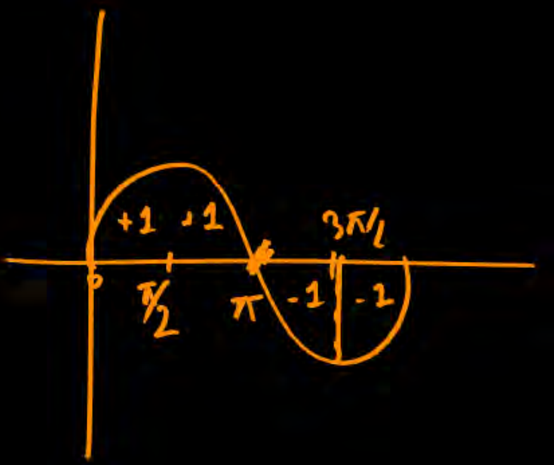
$$\int_0^{\pi} \sin(2x)\,dx = \left[\frac{-\cos(2x)}{2}\right]_0^{\pi}$$

$$= -\frac{1}{2}\left[\cos(2\pi) - \cos 0\right]$$

$$= -\frac{1}{2}\left[1 - 1\right]$$

$$= 0$$

$$\int_0^{3\pi/2} \sin x\,dx = +1 \checkmark$$



Question

copy (UD)

$$\int \left(\frac{x^3+2}{x^3}\right) dx =$$

$$\int \left(\frac{x^3}{x^3}+\frac{2}{x^3}\right) dx = \int \left(1+\frac{2}{x^3}\right) dx$$

$$= \int dx + 2\int \frac{1}{x^3} dx$$

$$= x + 2\left(\frac{x^{-3+1}}{-3+1}\right)$$

$$= x + 2\left(\frac{x^{-2}}{-2}\right)$$

$$= \left(x - \frac{1}{x^2}\right)$$ ✓

(0)

$$F = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

and $$dr = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$$

find $$W = \int \vec{F} \cdot d\vec{r}$$

$$W = \int \left(F_x \hat{i} + F_y \hat{j} + F_z \hat{k}\right) \cdot \left[dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}\right]$$

$\hat{i}\cdot\hat{i} = 1$

$\hat{i}\cdot\hat{j} = 0$

$\hat{i}\cdot\hat{k} = 0$

$$W = \int_{x_1}^{x_2} (F_x\,dx) + \int_{y_1}^{y_2} F_y\,dy + \int_{z_1}^{z_2} F_z\,dz$$

⊕ $\int \sin^2\theta \, d\theta = \int (\sin\theta)^2 \, d\theta$

$= \frac{(\sin\theta)^3}{3 \times 1} = \frac{\sin^3\theta}{3}$ ✗

$\int \frac{1 - \cos(2\theta)}{2} \, d\theta$

$= \int \frac{1}{2} \, d\theta - \int \frac{\cos(2\theta)}{2} \, d\theta$

$= \frac{1}{2}\theta - \frac{1}{2}\frac{\sin(2\theta)}{\times 2}$

$= \frac{\theta}{2} - \frac{\sin(2\theta)}{4}$ ✓

$\int \sin^2\theta \, d\theta =$

$= \int (1 - \cos^2\theta) \, d\theta$

We Know

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\cos(2\theta) = 1 - 2\sin^2\theta$

$2\sin^2\theta = 1 - \cos(2\theta)$

$\sin^2\theta = \frac{1 - \cos(2\theta)}{2}$

$$\int \cos^2\theta \, d\theta = \int \left[\frac{1+\cos(2\theta)}{2}\right] d\theta$$

$$= \int \frac{1}{2} d\theta + \int \frac{\cos(2\theta)}{2} d\theta$$

$$= \frac{\theta}{2} + \frac{\sin(2\theta)}{4}$$

(6) If gas is expanding from volume V_1 to V_2 under Isothermal Process then find work done.

$PV = nRT$ $\quad [dw = PdV]$ use

$P = \frac{nRT}{V}$

Soln

$$\int dw = \int_{V_1}^{V_2} P\,dV$$

put $P = \frac{nRT}{V}$ (isothermal $T = \text{cut}^n$, $P = \text{varible}$)

$$\text{Work} = \int_{V_1}^{V_2} \frac{nRT}{V}\,dV = nRT \int_{V_1}^{V_2} \frac{1}{V}\,dV = nRT \left[\log_e V\right]_{V_1}^{V_2}$$

$$= nRT \left[\log_e V_2 - \log_e V_1\right] = nRT \log_e \left(\frac{V_2}{V_1}\right)$$

Taking Average value

$$\langle \boxed{Y} \rangle_{\text{time-Avg}} = \frac{\int \boxed{Y}\, dt}{\int dt}$$

$$\langle \boxed{E} \rangle_{E\,Avg.} = \frac{\int E\, dt}{\int dt}$$

$$\langle P \rangle_{Avg} = \frac{\int P\, dt}{\int dt}$$

* Aiims

Space Avg value

$$\langle \square \rangle = \frac{\int \square\, dx}{\int dx}$$

Question

Finding of average value of continuous variable.

$$< L >_{\text{avg}} = \frac{\int L\, dt}{\int dt}$$

Value of length is discrete

$$L_{\text{avg}} = \frac{L_1 + L_2 + L_3 + L_4}{4}$$

$$L_{\text{avg}} = \frac{L_1 + L_2 + L_3 + \cdots + L_n}{n}$$

Question

If velocity of object $V = 2t + 1$ then find average velocity in 2 sec. ($t_i = 0$ to $t_f = 2$ sec)

Put ✓

$$\langle v \rangle = \frac{\int v\,dt}{\int dt} = \frac{\int_0^2 (2t+1)\,dt}{\int_0^2 dt} = \frac{\left(t^2 + t\right)_0^2}{(t)_0^2} = \frac{(2)^2+2) - 0}{2-0}$$

$$= \frac{4+2}{2} = \frac{6}{2}$$

$$= 3\ m/s$$

MR Scam

$$= \left(\frac{t^2+t}{t}\right)_0^2 = t+1\Big|_0^2 = (t+1)_0^2 = 3 - 1 = 2$$

Concept of Power

$1^2 = 1$

$2^2 = 4$

$2^3 = 2 \times 2 \times 2 = 8$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1024$

$3^2 = 9$

$3^3 = 27$

$3^4 = 81$

$3^5 = 243$

$3^6 = 243 \times 3$

$3^7 = 243 \times 9$

$4^2 = 16$

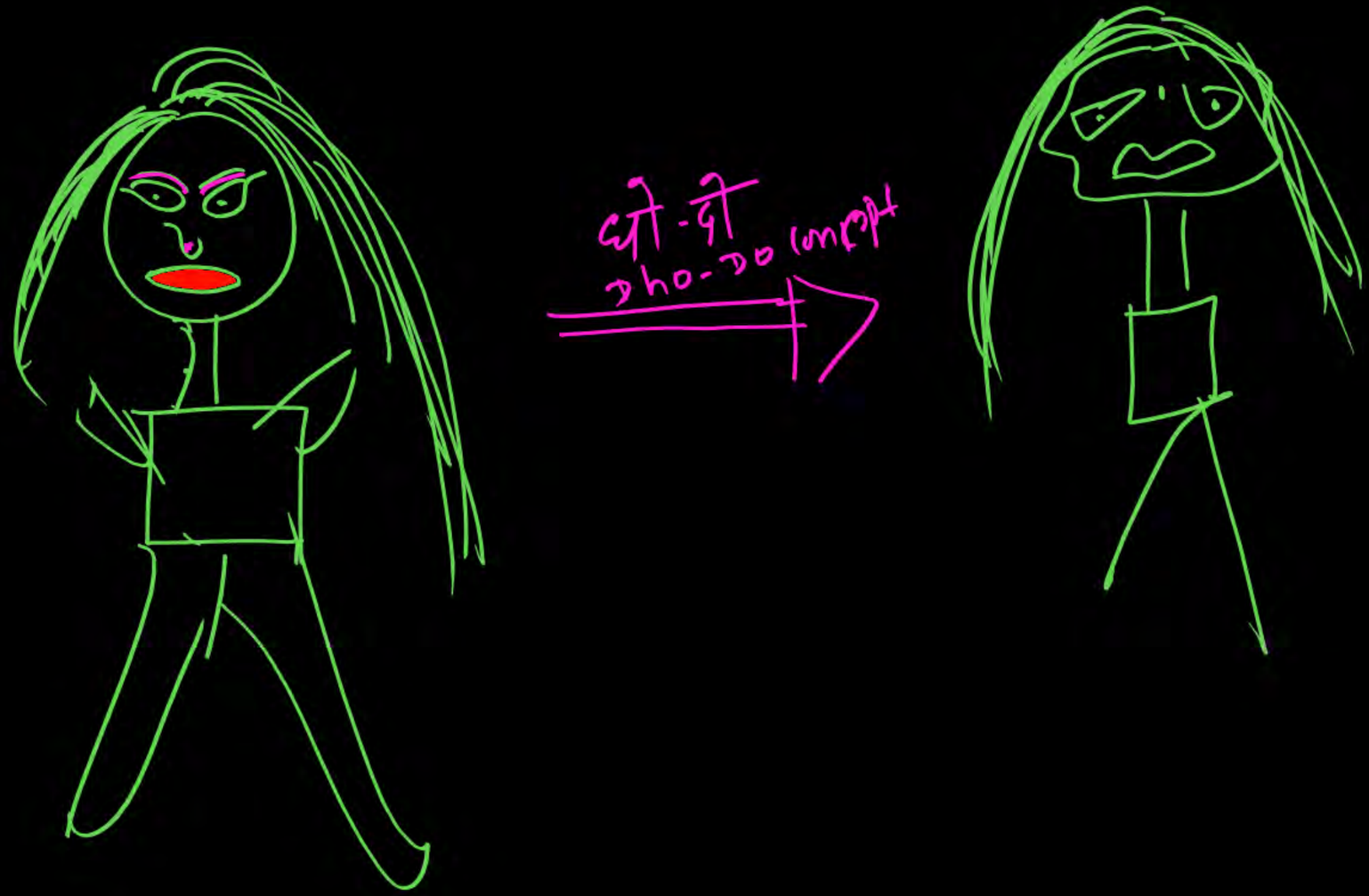
$4^3 = 64$

$4^4 = 4^2 \times 4^2 = 16 \times 16 = 256$

$10^2 = 100$

$10^3 = 1000$

$10^4 = 10000$


धो-दो
Dho-Do concept

log :- log ko पढ़ते kaise हैं ??

$\log b^{a} = \log (b)$ ka Power a ✗

$\log_b a = \log (a)$ on the base b ✓

$\log a^{c} = \log c$ on the base (a)

$\log y^{x} = \log (x)$ on the base (y)

Makeup walid lafz → Konsa of Pura

log → concept of Powz

Result

$\log_b a = ??\ x$ (let)

b → Base

x → Powz

..b (base) ka Powz kya lu (x)

ki Result a aa-Jay

dho-do

$b^x = a$

① $\log_2 8 = ??\ (x)$

↑ Base

$2^x = 8$

$x = 3$

log format

$\log_{y}^{x} = z$

Result

Base

Power

Base(Y) ka Power(z) equal to Resul(x)

→ dho-do compt

$y^{z} = x$

Powz

(2)

$\log_{10}^{100} = ??$ x (let)

(Base)

$10^{x} = 100$

$x = 2$

(3) $\log_{3}^{27} = ??$ x (let)

Base

$3^x = 27$

$x = 3$

(4) $\log_{100}^{10} = ??\ (x)$

(Base)

$100^x = 10$

$(100)^{1/2} = \sqrt{100} = 10$

$x = \frac{1}{2}$

(5) $\log_{2}^{128} = ??$

$2^x = 128$

$2^7 = 128$

$x = 7$

(6) $\log_{5}^{25} = ??\ (x)$

$5^x = 25$

$x = 2$ ✓

(7)

$$\log_4 x = 3$$

find x = ??

dho do concept

$$4^3 = x$$

$$\boxed{x = 64}$$

(8)

$$\log_3 y = 5$$

(8) find 'y'

$$3^5 = y$$

$$3 \times 3 \times 3 \times 3 \times 3 = y$$

$$\boxed{y = 243}$$

(9) $\log x^{100} = 2$

find x

$(x)^2 = 100$

$x = \sqrt{100} = 10$ ✓

$\log_{10} y^x = 2$

$y^2 = x$

(10) $\log_{(x)}^{216} = 3$

find x.

$x^3 = 216$

$x = (216)^{1/3}$

$= 6^{3 \times \frac{1}{3}} = 6$

log

Common log

$$\log_{10} x = \log x$$

Base missing ho to Base 10 lena hai. ✓

$$\Rightarrow \log_{10} 2 = 0.30$$

$$\Rightarrow \log_{10} 3 = 0.48$$

MR BOX

$$\log_e x = 2.303 \times \log_{10} x$$

Natural log

$$\log_e x = \text{Natural log} = \ln x$$

$\ln x$ Likha hai to base e lena hai.

(11) $\log e^2 = 2.303 \times \log 10^2$

$= 2.303 \ [0.30]$

$= 0.693$

(12) find $\log e^3$

$\log e^3 = \log 10^3 \times 2.303$

$= 0.48 \times 2.303$

$\cong 1.1$

Rule-1

\# $\log 2^2 = ??(x)$

$2^x = 2$

$\boxed{x = 1}$

$\log_{10} 10^{10} = 1$

$\log_5 5^5 = 1$

$\log_e e^e = 1$

$\log_3 3^3 = 1$

$\boxed{\log_a a^a = 1}$

Rule-2

$\log e^1 = ??\ (x)$

$e^x = 1$

$\boxed{x = 0}$

$8^0 = 1$

$2^0 = 1$

$e^0 = 1$

$\log e^1 = 0$

$\log 10^1 = 0$

$\log 5^1 = 0$

Rule-3

$\log_e (x)^n$ = log x to the Power n on the base e'

$$\log_e (x)^n = n \log_e x$$

(13) find value

$$\log_2 64 = ??\ (x)$$

Pure concept

$2^x = 64$

$\Rightarrow x = 6$ ✓

Rule-3

$$\log_2 (2)^6 = 6 \log_2 2 = 6$$

(14) find value

$$\log_7 49 = ??$$

$$\log_7 (7)^2 = 2 \log_7 7 = 2$$

Concept

$7^x = 49$

$x = 2$

Rule-4

$$\log e^{(xy)} = ??$$

$$= \log e^{x} + \log e^{y}$$

Concept of Power

$$10^{2} \times 10^{3} = 10^{5}$$

$$10^{2} \times 10^{3} = 10^{6}$$ ✗

Rule- 5

(#) $$\log e^{\left(\frac{x}{y}\right)} = \log e^{x} - \log e^{y}$$

$$\frac{10^{7}}{10^{3}} = 10^{7-3} = 10^{4}$$ ✓

(15) $\log_{10}^{6} = ??$

$\log_{10}^{(3\times2)} = \log_{10}^{3} + \log_{10}^{2}$

$= 0.48 + 0.30$

$= \underline{0.78}$

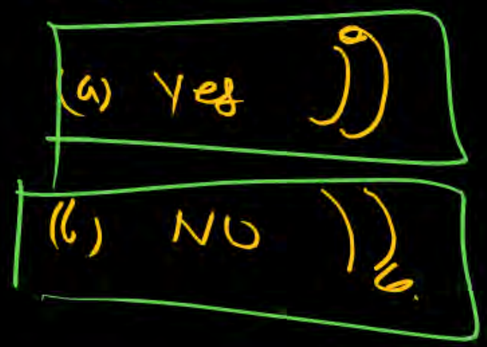
(16) find $\log_{10}^{(16/4)}$

$\log_{10}^{3/2} = \log_{10}^{3} - \log_{10}^{2}$

$= 0.48 - 0.30$

$= \underline{0.18}$

Kitne logo ko anti-log me Prob^n hai



Rule -6

(6) If $\log z^x = y$

↓

~~By taking anti-log~~ (dhoo-do concept)

$z^{(y)} = x$

$\log e^x = y$

↓

anti log

$e^y = x$

$\log_{10} y = x$

↳ anti-log → $10^x = y$

Rule-7

$$\log_{(e)} n^{x} = \frac{1}{n} \log_e x$$

$$\log_e^{(x)} n = n \log_e x$$

find value

(17)

$$\log 27^{3} = ??$$

→ Pure Impt ✓✓

Rule-7

$$\log_{(3)^3}^{3} = \frac{1}{3} \log_3^3 = \frac{1}{3}$$

$$27^{x} = 3$$

$x = \frac{1}{3}$

Rule-8

$$\log_y x = \frac{1}{\log_x y}$$

Rule-9

$$\log_y x \times \log_x y = 1$$

Question

18

$\log e^{25} + \text{Log}\, e^{4} - \log e^{10} =$ $\log_e e^{\left(\frac{25\times4}{10}\right)} = \log_e e^{\left(\frac{100}{10}\right)} = \log_e e^{10} = 2.303 \log_{10} 10^{10}$

$= 2.303$ ✓

Question

(15)

Find value of given expression:

$\log_{10}(4 \times 10^{-4})$ = ??

$$\log_{10}(4\times10^{-4}) = \log_{10}4 + \log_{10}(10^{-4})$$

$$= 2\log_{10}2 - 4\log_{10}10$$

$$= 2\times0.30 - 4\times1$$

$$= 0.60 - 4$$

$$= -3.4$$

Question

20

Find value of x In the given expression.

(i) $\log_2 x = -4$

$2^{-4} = x$

$\frac{1}{2^4} = x$

$x = \frac{1}{16}$

(ii) $\log_3 (x)^2 = 18$

$2 \log_3 x = 18$

$\log_3 x = 9$

$3^9 = x$

(21) find value of given log :—

- $\log_{10}(\sin\theta \cdot \operatorname{cosec}\theta) = 0$
- $\log_{10} 25 + \log_{10} 40 = \log_{10}(25\times 40) = \log_{10} 1000 = 3\log_{10} 10 = 3$
- $\log_{10} 200 - \log_{10} 2 = \log_{10}\left(\frac{200}{2}\right) = \log_{10} 10^2 = 2\times 1$
- $\log_{10} 200 =$

(22)

$\log_{16} 32 = ??\ (n)$

$\log_{2^4} (2)^5 = \frac{5}{4} \log_2 2$

$= \left(\frac{5}{4}\right)$ ✓

(23) find

$\log_{81} 27 = ??$

$\log_{(3)^4} (3)^3 = \frac{3}{4} \log_3 3 = \frac{3}{4}$

$= 0.75$

H/W

- $\log_8 16 =$
- $\log_{27} 3$
- $\log_{100} 1000 =$
- $\log_{10} (0.0001) =$
- $\text{Log}_{0.01} 10 =$

Question

H/w must do copy

Loudness of sound at a point is 50 dB then find intensity at that point if $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB where $I_0 = 10^{-12}$ w/m^2

1. 10^{-7} w/m^2
2. 10^{-5} w/m^2
3. 10^{8} w/m^2
4. 10^{7} w/m^2

Question

If current flowing through wire is I = 20 $e^{-\lambda t}$ find the time when current becomes 10 Amp. Where λ is 2.303.

H/W

Sanghani Assignment

must do

Question

Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

1. $y = \frac{x}{4} + 1$
2. $y = 3x - 2$
3. $y = 5x + 7$
4. $y = 4x + 8$

Question

The differentiation of function $f(x) = (3x + 2)^{3/2}$ w.r.t. x is

1. $\frac{9}{2}(3x + 2)^{1/2}$
2. $\frac{3}{2}(3x + 2)^{1/2}$
3. $\frac{3}{2}(3x + 2)^{-1/2}$
4. $\frac{9}{2}(3x + 2)^{-3/2}$

Question

A boy completes one round of a circular track of radius 20m in 50 seconds. The displacement at the end of 4 minute 10 second will be

1. 40 m
2. 20 m
3. 80 π m
4. Zero

Question

The derivative of function $f(x) = \log_e (2x)$ w.r.t. t is

1. $f'(x) = \frac{1}{2x}\frac{dx}{dt}$
2. $f'(x) = \frac{1}{x}\frac{dx}{dt}$
3. $f'(x) = \frac{1}{x^2}\frac{dx}{dt}$
4. $f'(x) = \frac{1}{2x^2}\frac{dx}{dt}$

Question

If $s = \frac{t}{2t+1}$ then find $\frac{ds}{dt}$

1. $\frac{5}{(4t+2)^2}$
2. $\frac{1}{(2t+1)^2}$
3. $\frac{1}{(3t-2)^2}$
4. $\frac{4}{(2t+1)^3}$

Question

If $x = 2t^3$ and $y = 3t^2$, then value of $\frac{dy}{dx}$ is

1. $\frac{t^2}{2}$
2. $\frac{2}{t^2}$
3. $\frac{1}{t}$
4. $\frac{1}{t^2}$

Question

If $x = 2(\theta + \sin\theta)$ and $y = 2(1 - \cos\theta)$, then value of $\frac{dy}{dx}$ is

1. $\tan\left(\frac{\theta}{2}\right)$
2. $\cot\left(\frac{\theta}{2}\right)$
3. $\sin\left(\frac{\theta}{2}\right)$
4. $\cos\left(\frac{\theta}{2}\right)$

Question

Find the slope of the tangent to the graph $y = x^3 - x^2$ at $x = -1$

1. 4
2. 5
3. 3
4. 6

Question

The displacement x of a particle moving along x-axis at time t is given by $x^2 = 2t^2 + 6t$. The velocity at any time t is:

1. $\frac{4t + 6}{x}$
2. $\frac{2t - x}{x}$
3. $4t + 6$
4. $\frac{2t + 3}{x}$

Question

The position of a particle with respect to time t along y-axis is given by: $y = 12t^2 - 2t^3$, where, y is in metres and t is in seconds. When the particle achieves maximum speed, the position of the particle would be

1. 64 m
2. 40 m
3. 16 m
4. 32 m

Question

$\int_2^4 \frac{1}{x}\,dx$ is equal to

1. $\ln 2$
2. $2 \ln 2$
3. $\ln\left(\frac{3}{2}\right)$
4. $2 \ln 4$

Question

If $y = \log_{10} x$, then the value of $\frac{dy}{dx}$ is

1. $\frac{1}{x}$
2. $\frac{x}{\log_e 10}$
3. $\frac{\log_{10} e}{x}$
4. $\frac{\log_e 10}{x}$

Question

Let $y = x^2 + x$, the minimum value of y is

1. $-1/4$
2. $1/2$
3. $1/4$
4. $-1/2$

Question

If $y = A \sin(\omega t - kx)$, then the value of $\frac{dy}{dx}$ is

1. $A \cos(\omega t - kx)$
2. $-A\omega \cos(\omega t - kx)$
3. $Ak \cos(\omega t - kx)$
4. $-Ak \cos(\omega t - kx)$

Question

$\int_{\infty}^{r} \frac{GM_1m_2}{r^2} dr$ is equal to

1. $\frac{Gm_1m_2}{r}$
2. $-\frac{Gm_1m_2}{r^3}$
3. $\frac{Gm_1m_2}{r^3}$
4. $-\frac{Gm_1m_2}{r}$

Question

If $y = A \sin(\omega t - kx)$, then the value of $\frac{d^2y}{dt^2} / \frac{d^2y}{dx^2}$ is

1. $\frac{k^2}{\omega^2}$
2. $-\frac{k^2}{\omega^2}$
3. $\frac{\omega^2}{k^2}$
4. $-\frac{\omega^2}{k^2}$

Question

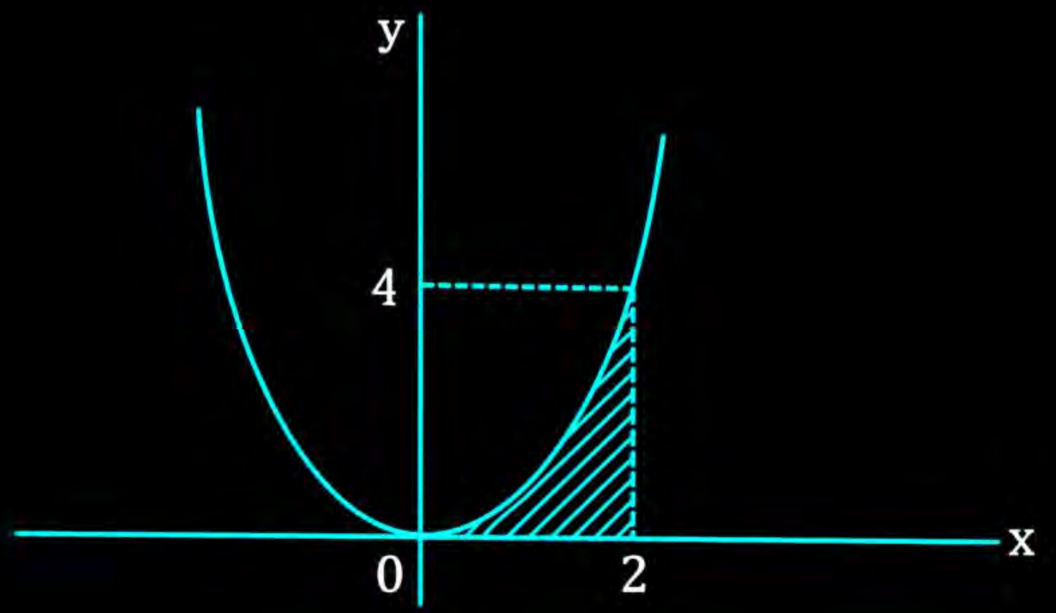
$$\int_0^L \frac{dx}{(ax+b)} =$$

1. $\frac{1}{b}\ln\left(\frac{aL+b}{b}\right)$
2. $-\frac{1}{a}\ln\left(\frac{aL+b}{b}\right)$
3. $\frac{1}{a}\ln\left(\frac{aL+b}{b}\right)$
4. $-\frac{1}{b}\ln\left(\frac{aL+b}{b}\right)$

Question

Consider the parabola $y = x^2$. The shaded area is

1. 2/3
2. 5/3
3. 7/3
4. 8/3



Question

A particle moves in a straight line, so that $x = \sqrt{t}$ then its acceleration is proportional to

1. Velocity
2. $(\text{Velocity})^{3/2}$
3. $(\text{Velocity})^{3}$
4. $(\text{Velocity})^{2}$

Question

Evaluate $\int x^2 . \sin(x^3) . dx$

1. $\frac{\sin x^3}{3} + c$
2. $\cos x^3 + c$
3. $\sin x^3 + c$
4. $\frac{-\cos x^3}{3} + c$

Question

If $F = \frac{2}{\sin\theta + \cos\theta}$ then the minimum value of F out of the following option is

1. 1
2. $\frac{1}{\sqrt{2}}$
3. $\frac{3}{\sqrt{2}}$
4. $\sqrt{2}$

Question

The acceleration of a particle moving along a straight line at any time t is given by $a = 4 - 2v$, where v is the speed of particle at any time t. The maximum velocity is

1. 4 m/s
2. 2 m/s
3. 6 m/s
4. Infinity

Thank You